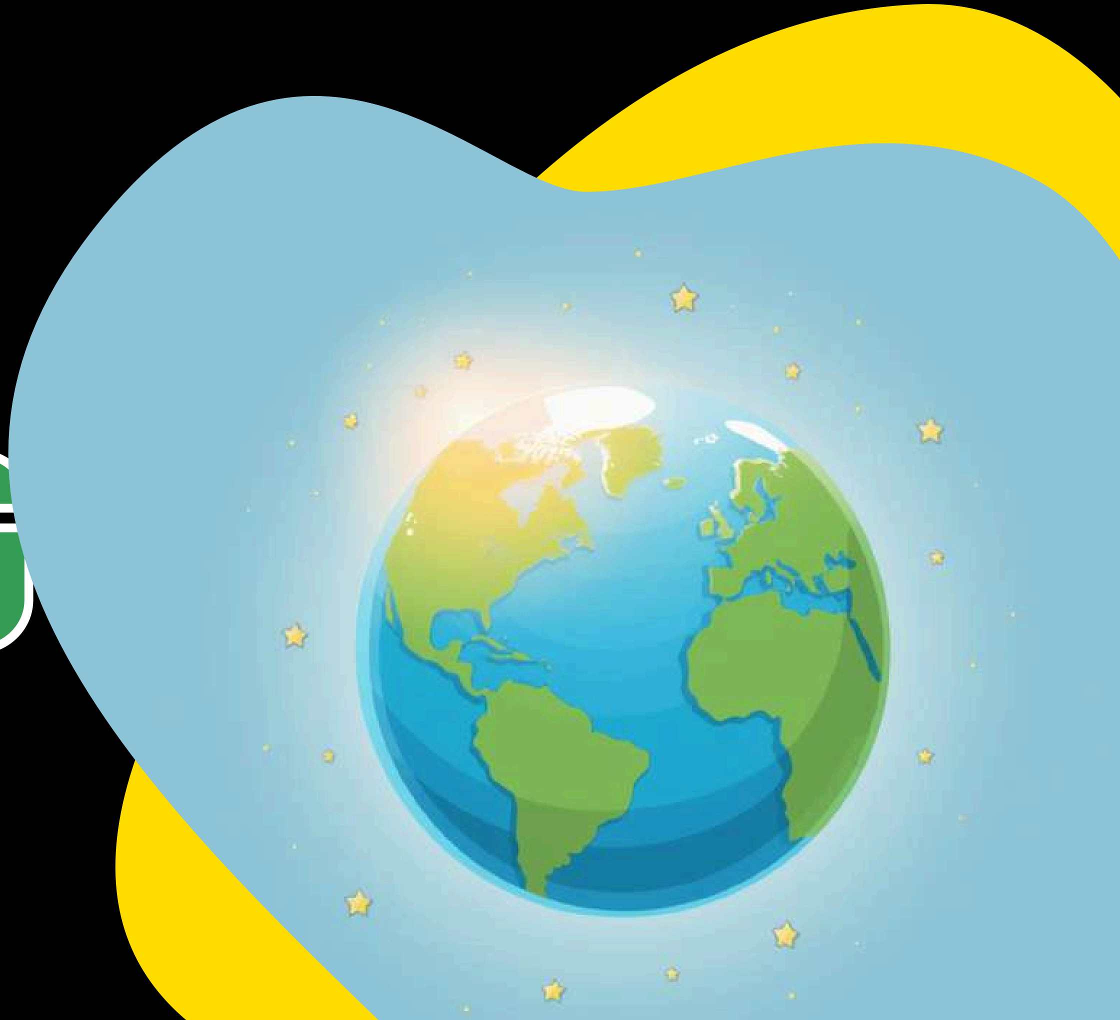
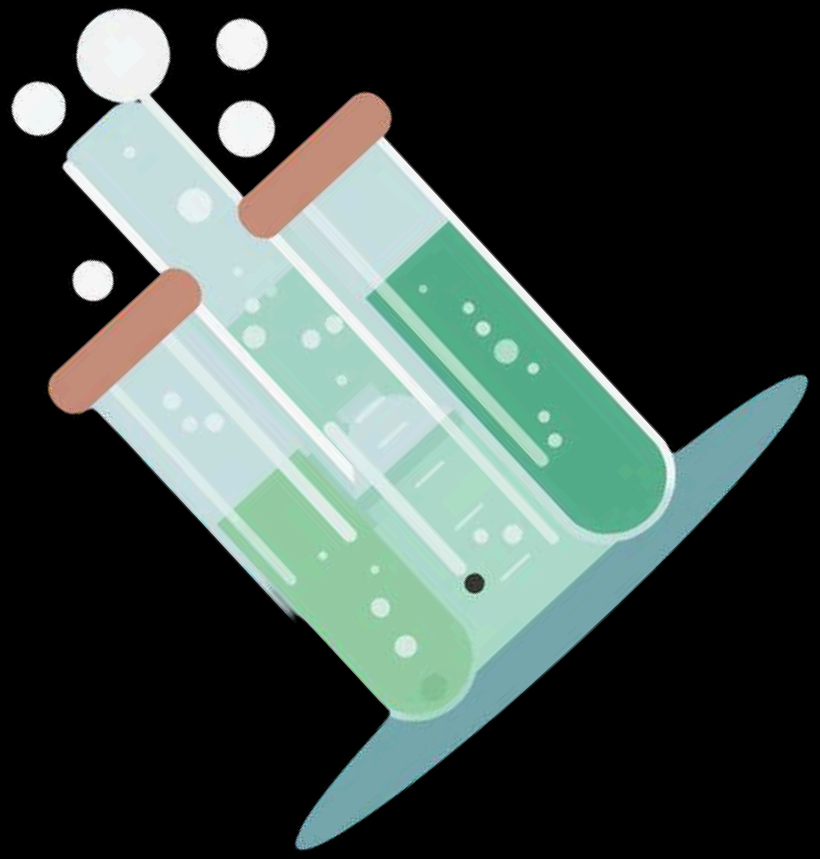
SOLUTIONS TO HELP OUR PLANET

CLIMATE ENGINEERING



Climate change is making Earth warmer and changing weather patterns. Key fact: Earth's temperature has risen by about 1.1°C since 1880!

WHY DOES OUR PLANET NEED HELP?



WHAT IS CLIMATE ENGINEERING?

Climate engineering means using science to fix or slow down climate change. It includes amazing ideas and technologies to protect Earth's atmosphere!

WHY DO WE NEED CLIMATE ENGINEERING?

Our planet needs extra help because natural processes alone cannot keep up with the changes happening to our climate.

Greenhouse gases trap heat, causing global warming
Nature alone can't remove carbon fast enough
We need new ways to protect plants, animals, and people
Helps buy time while we reduce pollution

TWO MAIN TYPES OF CLIMATE ENGINEERING

Solar Radiation Management
Reflects sunlight back into space to cool Earth down. Uses mirrors, clouds, or white surfaces to bounce heat away.

Carbon Dioxide Removal
Takes CO_2 out of the air using trees, machines, or storing it underground safely.

Reflect sunlight back into space using giant mirrors or tiny aerosol particles in the atmosphere.
Brighten clouds by spraying sea salt to make them whiter and more reflective, helping cool the Earth.

SOLAR RADIATION MANAGEMENT (SRM)

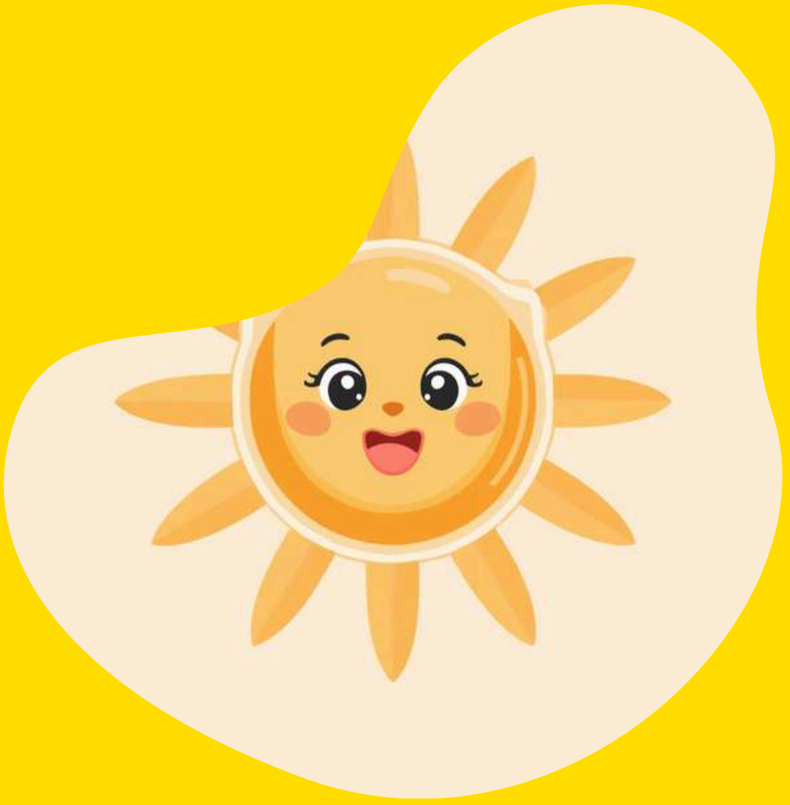
Planting trees and forests absorbs CO_2 naturally as plants grow.
Using machines that capture CO_2 from air and storing CO_2 underground safely helps reduce greenhouse gases.

CARBON DIOXIDE REMOVAL (CDR)

HOW DO THESE SOLUTIONS WORK?

Solar Radiation Management reflects sunlight back into space to cool Earth, while Carbon Dioxide Removal captures and stores greenhouse gases underground or in plants.

PROS AND CONS OF CLIMATE ENGINEERING



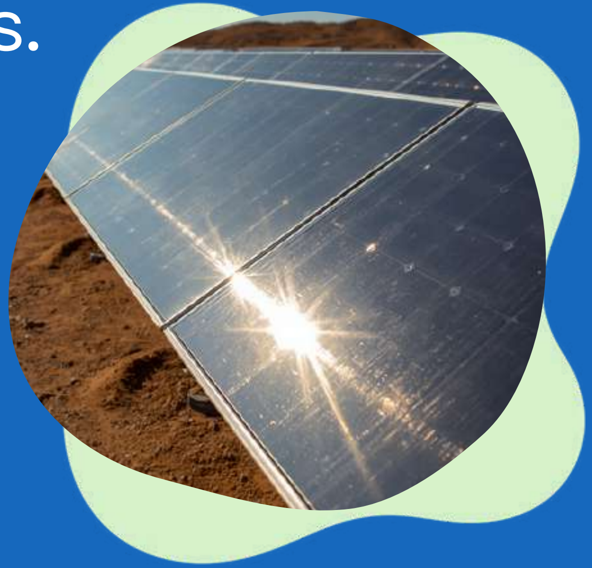
Climate engineering has benefits and risks. Let's explore both sides!

- ✓ Quick results to cool Earth
- ✓ Helps protect ecosystems
- ✓ Supports other climate efforts
- ✓ Creates new technology



REAL-WORLD CLIMATE ENGINEERING PROJECTS

Around the world, scientists and communities are working together! Tree planting projects in India restore forests, while CO2 capture machines in Iceland pull carbon from the air. Solar reflectors help cool buildings in hot cities.



TIMELINE: CLIMATE ENGINEERING THROUGH HISTORY

1960S: FIRST IDEAS

Scientists first imagined ways to cool Earth by reflecting sunlight. These were just ideas on paper!

2000S: RESEARCH BEGINS

Researchers started testing carbon capture and cloud brightening in labs around the world.

2020S: REAL PROJECTS

Real climate engineering projects are now happening! Trees are being planted and CO_2 machines are running.

WHAT CAN KIDS DO TO HELP?

You have the power to make a difference! Here are simple ways you can help protect our planet every day.

Learn about climate change and solutions

Plant trees in your community

Save energy at home and school

Share what you learn with friends and family

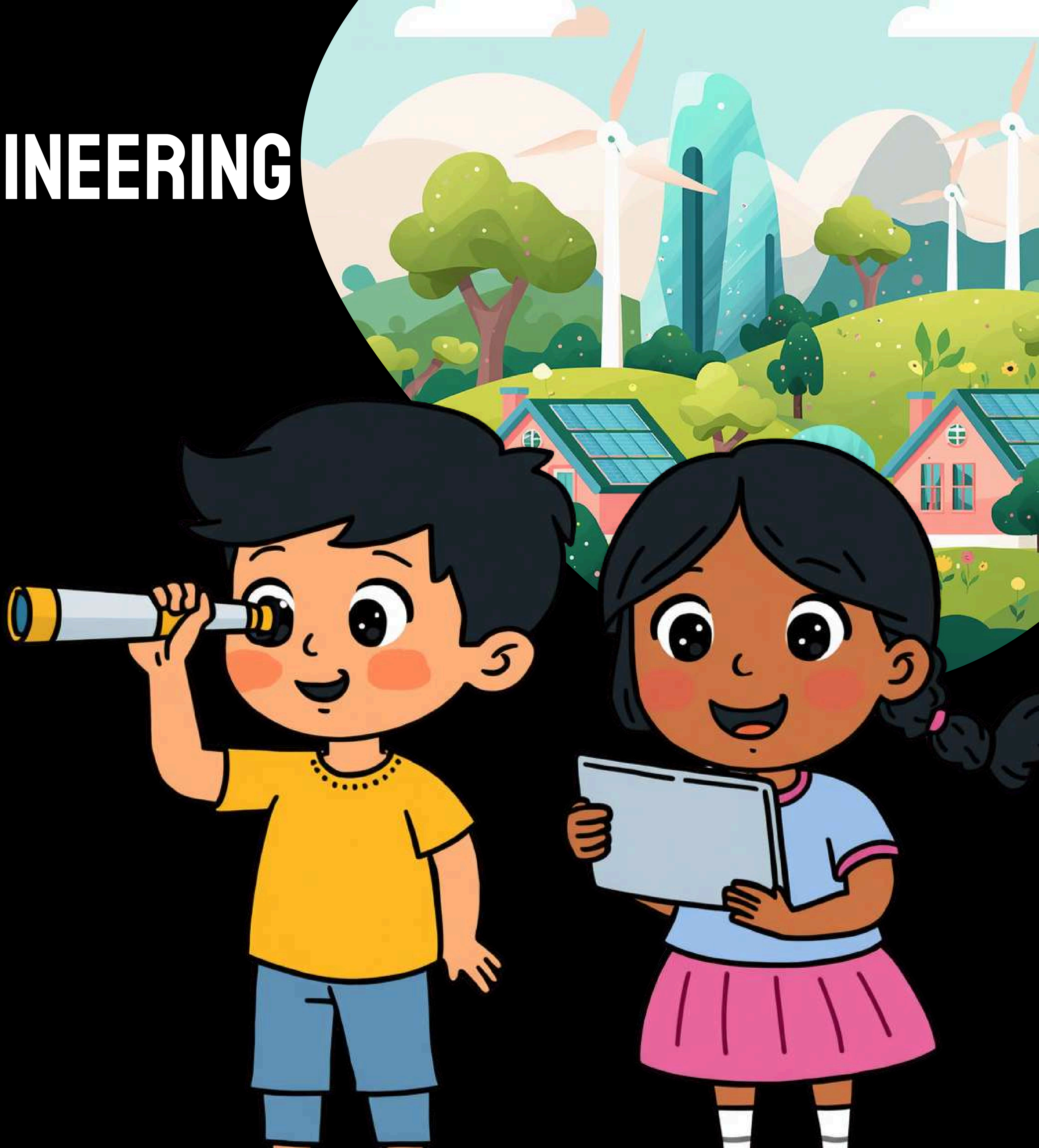
FUN FACT ABOUT CLIMATE ENGINEERING

"Science gives us superpowers to protect our planet!"

Every day, scientists around the world are inventing amazing new ways to help Earth stay cool and healthy!

THE FUTURE OF CLIMATE ENGINEERING

Scientists around the world are working on exciting new ideas every day to protect our planet! With curiosity and STEM learning, we all can help make Earth safe and healthy for future generations. Your ideas might change the world someday!



BONUS: QUIZ TIME!

Q1: What are the two main types of climate engineering?
Q2: Name one way to remove carbon dioxide from the air.

Q3: How does Solar Radiation Management help cool Earth?
Test your knowledge and see how much you learned!

THANK YOU

GRADE 6 STEM

THANK YOU FOR EXPLORING CLIMATE ENGINEERING WITH US! KEEP BEING CURIOUS ABOUT SCIENCE AND OUR PLANET.

FOR LEARNING!